

Zhicheng ZHANG — CV

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“We Must Know, We Will Know.” — David Hilbert

Z. Zhang is currently a Specifically Appointed Researcher at The University of Osaka, Suita, Japan. Before he joined the UOsaka, he was a program researcher at Kyoto University. He received Ph.D. degree in Informatics from Osaka University, Suita, Japan, in 2024, and the M.Sc. degree in Mathematics from Guilin University of Electronic Technology, Guilin, China, in 2020, and B.A. degree in Japanese from Changzhou Institute of Technology, Changzhou, China, in 2017. From 2019 to 2020, he has been a visiting researcher at EECS at The University of Kitakyushu, Fukuoka, Japan.

His research interests include *Applied Koopmanism, Dynamical Systems, Sparse Modeling, Robust Optimization, Control Theory and its Applications*. He is a member of SICE, IEEE, SIAM and INFORMS.

Position and Experience

The University of Osaka Specifically Appointed Researcher Fields: Nonlinear dynamical systems Promotor: Prof. Sho Shirasaka	Osaka, Japan 206.10 – present
Kyoto University Specific Program Researcher Fields: Applied Koopmanism to Weather Systems Advisor: Prof. Yoshihiko Susuki	Kyoto, Japan 2024.04 – 2026.09
Osaka University (now The University of Osaka) Dept. of Information and Physical Sciences Research Assistant (RA) Teaching Assistant (TA)	Osaka, Japan 2020.10 – 2024.03
The University of Kitakyushu Research Fellow (Visiting Researcher -Master Student), EECS Fields: Sparse Modeling and Optimal Control Host Advisor: Prof. Masaaki Nagahara	Fukuoka, Japan 2019.09 – 2020.09

Education

Osaka University (now The University of Osaka) Ph.D., Information Science and Technology Fields: Probabilistic Robustness for Sparse Control Supervisor: Prof. Yasumasa Fujisaki	Osaka, Japan 2020.10 – 2024.03
Guilin University of Electronic Technology M.S., Mathematics, Fields: Dynamical Systems and Complex Networks Advisor: Prof. Zhongjun Ma	Guilin, China 2017.09 – 2020.06
Changzhou Institute of Technology B.A., Japanese (Major) B.S., Applied Mathematics (Minor)	Changzhou, China 2013.09 – 2017.06

Research Interests (Operator $\times$ Engineering)

- Nonlinear Dynamical System, Applied Koopman Operator, Data-Driven Computational Methods
Sparse Modeling, Robust Optimization and its Control Applications, Large-scale Networked Systems

Honors and Awards

- SICE Annual Conference Young Author's Award (YAA) Finalist, SICE-FES'25, SICE, Sep., 2025
- 2020 Guangxi Outstanding Graduates, China, 2020 (top 1%)
- Outstanding Master's Thesis of GUET, China, 2020 (top 3%)
- Postgraduate Scholarship, First Prize, GUET, China, 2019, 2020 (top 3%)
- Graduate Fellowship for Study Abroad, GUET, China, 2019
- National Scholarship for Master's Student, China, 2019 (top 3%)

Research Projects and Academic Foundations

Towards Digital Twins to Controlled Cyber-Physical Networked Systems

Principal Investigator (PI), FY-2026 First-Term International Exchange Grant 2026.04 – 2026.09

- Tateisi Science and Technology Foundation, Grant No. XXX (JPY 500,000 Yen)

Dimension Reduction Using Koopman Mode Decomposition

Co-Collaborator, Kyoto University 2024.04 – 2027.3

- JST Moonshot Program Goal 8, Grant No. JPMJMS2389-3-2

Partial Component Synchronization of Nonlinear Networks and its Applications

Collaborator, Guilin University of Electronic Technology 2019.01 – 2021.12

- Guangxi Natural Science Foundation, China, Grant No. 2018GXNSFAA281068 (CNY 50,000)

Cooperative Control of Multi-Agent Networked Systems

Principal Investigator (PI), School of Mathematics and Computing Science, GUET 2019.09 – 2020.08

- The Study Abroad Program for Graduate Student of GUET, Grant No. GDYX2019015 (JPY 1,800,000 Yen)

Intermittent Feedback Control of Nonlinear Multi-Agent Systems

Principal Investigator (PI), School of Mathematics and Computing Science, GUET 2018.12 – 2020.06

- Cultivation of Excellent Thesis Project of GUET Graduate Education, Grant No. 2018YJSPY01 (CNY 10,000)

Positions of Responsibility

- **Teaching Assistant** for Undergraduate courses like *Mathematical Analysis*, *linear Algebra*, *matrix computation*, *control theory*, and *Calculus*.
- **Research Assistant** for Graduate courses like *operations research*, and *research seminars*.

Professional Service

Reviewer (Journals & Conferences)

- *International Journal of Robust and Nonlinear Control (IJRNC)*, *IEEE Transactions on Systems, Man and Cybernetics (IEEE TSMC)*, *IEEE Transactions on Automation Science and Engineering (IEEE TASE)*, *Physics of Fluids*, *CHAOS*, *Digital Communications and Networks*, *IFAC Symposium on Robust Control Design (ROCOND)*, *IFAC World Congress (WC)*, *European Control Conference (ECC)*, *IEEE Conference on Decision and Control (CDC)*, *IEEE Int. Conf. Advanced Robotics and Mechatronics (ICARM)*

Publications

Peer Reviewed Journals

- [J3] Z. Zhang, Y. Susuki and A. Okazaki, "Extracting transient Koopman modes from short-term weather simulations with sparsity-promoting dynamic mode decomposition," *Nonlinear Theory and Its Applications, IEICE*, 2026, vol. 17, no. 3, 1333-1361. ([J-Stage](#); [arXiv.2506.14083](#))
- [J2] Z. Zhang and Y. Fujisaki, "Sparse feedback controller: From open-loop solution to closed-loop realization," *SICE Journal of Control, Measurement, and System Integration*, 2023, vol., 16, no. 1, 286-296. ([Taylor & Francis](#); [arXiv.2303.15175](#))
- [J1] Z. Zhang, Z. Ma and Y. Wang, "Partial component consensus of leader-following multi-agent systems via intermittent pinning control," *Physica A: Statistical Mechanics and its Applications*, 2019, 536: 122569 ([Elsevier](#)).

Refereed International Conference Proceedings

- [C9] Z. Zhang, F. Fausto, Z. Ma, and M. Nagahara, "Towards lag consensus with noisy digital twins perception in second-order multi-agent cyber-physical systems," *23rd Proc. IFAC World Congress (IFAC WC'26)*, IFAC, Busan, Korea, TuB07.5, Aug. 22-28, 2026 (oral presentation; [arXiv.2605.04692](#))
- [C8] Z. Zhang, Y. Susuki and A. Okazaki, "Koopman mode decomposition of transient weather dynamics: A case study on humidity ratio data field," *2025 Int. Sympos. on Nonlinear Theory and Its Applications (NOLTA'25)*, IEICE, Okinawa, Japan, Oct. 27-31, 2025, pp. 1-4.
- [C7] Z. Zhang, Y. Susuki and A. Okazaki, "Sparsity-promoting dynamic mode decomposition applied to sea surface temperature fields," *Proc. SICE Festival with Annual Conference (SICE FES'25)*, Chiang Mai, Thailand, Sep. 9-12, 2025, pp. 657-662 ([IEEE](#); [arXiv.2507.05711](#))
Finalist of SICE Annual Conference Young Author's Award (YAA), SICE-FES 2025
- [C6] Z. Zhang and Y. Fujisaki, "Data-driven sparse feedback control with Schur- α stability," *SICE International Symposium on Control System (ISCS'24)*, SICE, Hiroshima, Mar., 2024, p. 3M1-2 ([preprint](#))
- [C5] Z. Zhang and Y. Fujisaki, "Risk assessment for sparse optimization with relaxation," *Proc. of the 55th Int. Sympos. on Stocha. Syst. Theory and its Appli. (SSS'23)*, ISCIE, Nov., 2024, pp. 20-23. ([J-Stage](#))
- [C4] Z. Zhang and Y. Fujisaki, "Risk-aware sparse predictive control", *Preprint of the 22nd IFAC Word Congress*, IFAC, Yokohama, July 2023, pp. 1477-1480. ([preprints](#))
- [C3] Z. Zhang and Y. Fujisaki, "Sparse feedback control realization using linear dynamic compensator," *SICE International Symposium on Control System (ISCS'23)*, SICE, Mar. 2023, p. 3M1-4.
- [C2] Z. Zhang and Y. Fujisaki, "Sparse robust control design via scenario optimization", *Proc. of the 53rd Int. Sympos. Stocha. Syst. Theory and Its Appli. (SSS'21)*, ISCIE, Oct., 2022, pp. 61-64 ([J-Stage](#))
- [C1] Z. Zhang and M. Nagahara, "Linear quadratic tracking control with sparsity-promoting regularization," *2021 American Control Conference (ACC'21)*, IEEE, May 2021, pp. 3812-3817 ([IEEE](#))

International Conference Presentations

- [ICP3] Z. Zhang and Y. Susuki, "Dual Koopman operator to large-scale ensemble weather dynamics," *International Symposium on Weather Controllability*, Tokyo, October 31, 2025 (Poster Session)
- [ICP2] Z. Zhang, Y. Susuki, and A. Okazaki, "Koopman analysis of large-scale transient simulation data on weather dynamics," *SIAM Conference on Applications of Dynamical Systems (SIAM DS'25)*, Denver, USA, May 11-15, 2025. (Oral Presentation)
- [ICP1] Z. Zhang and Y. Susuki, "Koopman analysis of weather dynamics using SCALE simulation data," *International Symposium on Weather Controllability*, Tokyo, October 6, 2024 (Poster Session)

Refereed Domestic Conference

- [Dom1] Z. Zhang, Y. Susuki, and A. Okazaki, "Exploring SCALE weather data via Koopman modes," *The 67th Japan Joint Automatic Control Conference*, Himeji, Nov., 2024, 11J-5, pp. 274-275. ([J-Stage](#))

Preprints

- [P1] Z. Zhang, W. Zhang, Z. Ma, "Wait-track consensus for nonlinear multi-agent system under control input failures," (Nonlin Dyna)

[P2] Z. Zhang, Z. Ma, and N. Masaaki, Y. Fujisaki, “Lyapunov Methods for Stabizing the Nonlinear Second-order agents,” (Submitted to IEEE Trans XX)

Thesis

[T1] Master’s Thesis: Consensus of Classes of Nonlinear Multi-agent Network Systems via Intermittent Control, *China National Knowledge Infrastructure (CNKI)*, June, 2020 (in Chinese)

[T2] Ph.D. Dissertation: Modeling, Robustness and Stability for Sparse Optimal Control of Dynamical Systems, *Osaka University Knowledge Archive (OUKA)*, March, 2024. (Doi: [10.18910/96217](https://doi.org/10.18910/96217))

Technical Strengths

- **Languages:** Mandarin Chinese (native), Japanese (N2), and English (fluent).
- **Skills:** $\text{\LaTeX}$, Matlab, Python, Julia, Multimodel Ensemble Prediction System (MEPS)
- **Certificate:** eAPRIN Learning, JLPT (N2)

Personal Information

- Born in Dec. 19, 1994, [Wuxi Prefecture](#), China
- ⊗ Citizenship: Chinese

Referees

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